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Digital Product Passport - System interoperability

Passeport numérique des produits - Interopérabilité du système

Digitaler Produktpass - System-Interoperabilität

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CEN-CENELEC Management Centre:
Rue de la Science 23, B-1040 Brussels

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Contents

Page

European foreword ... 3
Introduction ... 4
1 Scope ... 5
2 Normative references ... 5
3 Terms and definitions ... 5
4 Semantic interoperability ... 6
4.1 Semantic model ... 6
4.1.1 General ... 6
4.1.2 Semantic model of the digital product passport ... 6
4.2 Data model requirements to manage DPP changes ... 12
4.3 Model for repositories for the data dictionaries ... 12
5 Technical interoperability ... 13
5.1 General ... 13
5.2 Serialization syntax ... 13
5.2.1 General ... 13
5.2.2 Naming conventions and built-in types ... 13
5.2.3 JSON built-in types for values ... 13
5.2.4 Serialization of the DigitalProductPassport class ... 14
5.2.5 Serialization of a DataElementCollection ... 14
5.2.6 Serialization of a SingleValuedDataElement ... 15
5.2.7 Serialization of a MultiValuedDataElement ... 15
5.2.8 Serialization of a RelatedResource ... 16
5.2.9 Serialization of a MultiLanguageDataElement ... 16
Annex A (normative) Expanded full data model serialized in JSON ... 17
Annex B (normative) Data model compressed serialization in XML ... 21
Annex ZA (informative) Relationship between this European Standard and the essential requirements of (EU) Regulation 2024/1781 aimed to be covered ... 24
Bibliography ... 25

European foreword

This document (EN 18223:2026) has been prepared by Technical Committee CEN/CLC/JTC 24 “Digital product passport – Framework and system”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2026, and conflicting national standards shall be withdrawn at the latest by November 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

According to the CEN-CENELEC Internal Regulations, the national standards organisations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom.

Introduction

0.1 Background

A digital product passport (DPP) is a key enabling mechanism to make product information traceable and accessible across value chains - supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions and to support a circular economy. The implementation of digital product passports will be carried out progressively. Sector-specific initiatives will determine the precise DPP content and requirements for individual product groups.

To support the implementation of DPPs, 8 standards have been developed so far:

— EN 18219:2026 – Digital product passport – Unique identifiers

— EN 18220:2026 – Digital product passport – Data carriers

— EN 18216:2026 – Digital product passport – Data exchange protocols

— EN 18222:2026 – Digital Product Passport – Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability

— EN 18223:2026 – Digital Product Passport – System interoperability (this document)

— EN 18221:2026 – Digital product passport – data storage, archiving, and data persistence

— EN 18239:—[1] – Digital Product Passport – access rights management, information system security, and business confidentiality

— EN 18246:—[2] – Digital Product Passport – Data authentication, reliability and integrity

0.2 Organizational interoperability

Organizational interoperability covers the following aspects:

— easy integration into existing technology frameworks;

— human and machine (software) access.

0.3 Semantic interoperability

Semantic interoperability covers the following aspects:

— allow data structuring with standardized or established data schema languages;

— enable the usage and re-usage of known ontologies or context namespaces;

— avoid sector specific ontologies – allow parallel usage of different ontologies issued by the European Commission in delegated acts with a horizontal layer as general as possible (technology, ontology and domain agnostic).

0.4 Technical interoperability

Serialization of data with standardized data format.

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[1] Under preparation. Stage at the time of publication: prEN 18239:2025.
[2] Under preparation. Stage at the time of publication: prEN 18246:2025.

1 Scope

The scope of this document includes:

— the semantic description of a product, including its properties where relevant and the semantic aspects to represent the product lifecycle;

— a common information model allowing for the implementation of data dictionary systems;

— metadata models and formats to be used in exchange and representation, allowing for the integration of dictionaries;

— rules on how to systematically use such metadata models when developing product group specific data models and dictionaries;

— technical and organizational interoperability.

This document follows the approach of standard interoperability layers and proposes the following aspects in this regard.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 8601-1:2019,[3] *Date and time — Representations for information interchange — Part 1: Basic rules*

EN 18219, *Digital product passport - Unique identifiers*

EN 18221, *Digital product passport - data storage, archiving, and data persistence*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
interoperability
capability of two or more entities to exchange items in accordance with a set of rules and mechanisms implemented by an interface in each entity, in order to perform their specified tasks

[SOURCE: ISO 18435-1:2009 [1], 3.12]

[3] As impacted by ISO 8601-1:2019/A1:2022.

3.2
semantic interoperability
ability for data shared by systems to be understood at the level of fully defined domain concepts

[SOURCE: ISO 18308 [2], 3.45]

3.3
technical interoperability
ability that covers the applications and infrastructures linking systems and services

3.4
digital product passport
DPP
digital record of product characteristics throughout its life cycle

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability.

3.5
digital product passport id
DPP ID
unique identifier of an instance of the digital product passport

4 Semantic interoperability

4.1 Semantic model

4.1.1 General

This document uses UML class diagrams ISO/IEC 19505-2 [3] to illustrate the core concepts of a product and DPP, their data, properties and their relationships. A JSON representation is provided in Clause 5. If there are discrepancies between the UML diagrams, text and JSON representations the prose text of Clause 4, including the tables, is authoritative.

4.1.2 Semantic model of the digital product passport

4.1.2.1 Digital product passport

The model presented below in Figure 1 defines what the entities are that compose a digital product passport instance and the data points it is required to include.

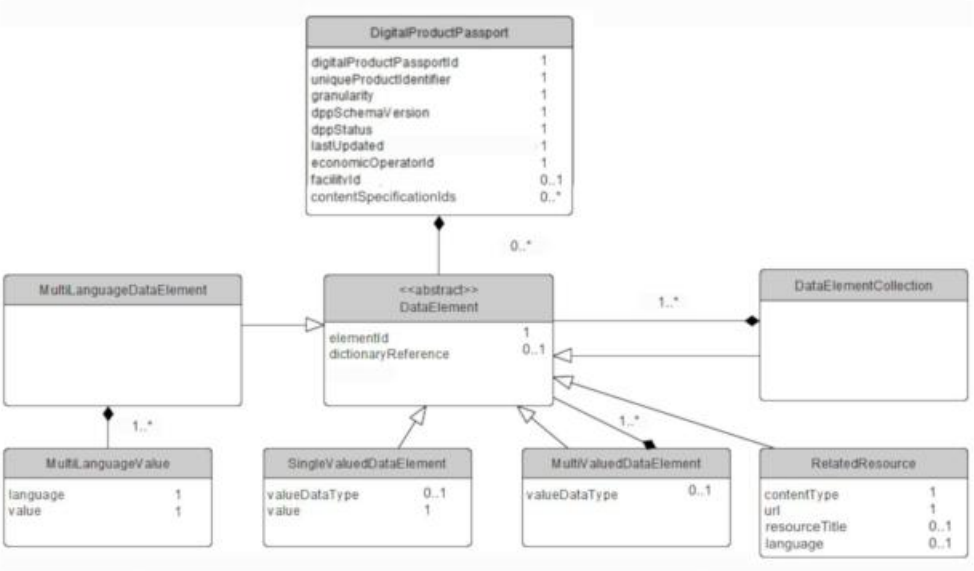


Figure 1 — UML representation of the entities of the semantic model of the DPP

Figure 1 presents the base semantic model of a digital product passport.

The digital product passport is modelled through the class *DigitalProductPassport* as a digital container that contains zero to many *DataElements.*

The model is structured based on the following principles:

— *DataElement* is an abstract class. It represents the generic concept of a data point. As an abstract class, it is the common superclass for all other classes, and it is never used directly in a DPP instance.

— *DataElementCollection*: A concrete subclass of *DataElement* (see 4.1.2.3) that acts as a container for a set of other named *DataElements*. This allows for creating complex structures with mixed data types, as it can contain *SingleValuedDataElements* (see 4.1.2.5), *MultiValuedDataElements* (see 4.1.2.6), *RelatedResource* (see 4.1.2.7), and other nested *DataElementCollections* (see 4.1.2.4).

— *SingleValueDataElement*: This is the simplest concrete subclass of *DataElement*. It represents a single unit of information (see 4.1.2.5).

— *MultiValuedDataElement:* A concrete subclass of *DataElement* that contains an ordered list of other *DataElements*, all of which shall be of the same type. For example, it can contain a list of *SingleValueDataElements* (like an array of numbers) or a list of *DataElementCollections* (like an array of similar objects) (see 4.1.2.4).

— Other Concrete subclasses: *MultiLanguageDataElement* (see 4.1.2.8) and *RelatedResource* (see 4.1.2.7) are other specialized subclasses of *DataElement*, used for representing a value in multiple languages or linking to an external resource, respectively.

The following data points (see Table 1) shall be published as attributes of a digital product passport instance:

Table 1 — Attributes of the DigitalProductPassport class

Attribute name	Description	DataType	Cardinality
digitalProductPassportId	unique identifier of an instance of the digital product passport	String	[1]
uniqueProductIdentifier	unique string of characters for the identification of a product, that also enables a web link to the digital product passport	String formatted according to EN 18219	[1]
granularity	level of granularity of the uniqueProductIdentifier	Enumeration according to 4.1.2.2	[1]
dppSchemaVersion	reference standard the DPP instance schema refers to	String	[1]
dppStatus	status of the DPP instance as digital resource	String	[1]
lastUpdated	date and time of the latest update to the DPP instance	String formatted as Timestamp UTC- based according to ISO 8601-1:2019	[1]
economicOperatorId	unique string of characters for the identification of an actor involved in a product's life cycle	String formatted according to EN 18219	[1]
facilityId	unique string of characters for the identification of locations or buildings involved in a product's value chain or used by actors involved in a product's life cycle	String formatted according to EN 18219	[0..1]
contentSpecificationIds	reference to horizontal or product-type related content specifications for the DPP	String	[0..*]
{DataElement}	data elements of DPP	DataElement (abstract) as per 4.1.2.3	[0..*]

EXAMPLE Example values for the "dppStatus" attribute are:

— active,

— inactive,

— archived,

— invalid

Any further enumeration value provided by relevant legal acts.

The digitalProductPassportId shall be globally unique and should be based on a URI/URL structure.

{DataElement} is a placeholder for additional content or other attributes of the DPP. The notation {<name>} means that the naming is not fixed. This includes data points mentioned in ESPR [4], Annex III and not included in Table 1.

4.1.2.2 Enumeration

Some of the attributes required by the DigitalProductPassport class are enumerations whose values are predetermined.

The values allowed for the "granularity" attribute are:

— model,

— batch,

— item.

4.1.2.3 DataElement

The DataElement is an abstract class which is the base class for its concrete subclasses: *DataElementCollection* (4.1.2.4) *SingleValuedDataElement* (4.1.2.5), *MultiValuedDataElement* (4.1.2.6), *RelatedResource* (4.1.2.7) and *MultiLanguageDataElement* (4.1.2.8). It specifies the following attributes (see Table 2).

Table 2 — Definition of DataElement attributes

Attribute Name	Description	Datatype	Cardinality
elementId	The relative identifier of the *DataElement* shall be unique within its location (i.e. in the DPP or in the level of *DataElementCollection* or *MultiValuedDataElement*) and it is recommended to be unique in the DPP instance. The corresponding absolute path of the data element shall be unique within the DPP instance.	String	[1]
dictionaryReference	The reference to the unique identifier of the data point specification defined in the repository/data dictionary	String	[0..1]

4.1.2.4 DataElementCollection

The *DataElementCollection* is a collection of *DataElements* included in the digital product passport entity. It contains at least one *DataElement*. This class inherits the attributes *elementId* and *dictionaryReference* from the abstract *DataElement* class.

The *DataElementCollection* is an object and as such it can contain semantically different objects. A *DataElementCollection* shall contain at least one instance of a *DataElement*'s subclasses (*DataElementCollection*, *SingleValuedDataElement*, *MultiValuedDataElement* or *RelatedResource*).

4.1.2.5 SingleValuedDataElement

The *SingleValuedDataElement* represents a single data point with one value. This class inherits the attributes *elementId* and *dictionaryReference* from the abstract *DataElement* class and adds the following specific attributes (see Table 3).

Table 3 — Definition of *SingleValuedDataElement* attributes

Name	Description	Datatype	Cardinality	Example
value	Contains the data of the value encapsulated in the DataElement	< any JSON datatype > as described in Clause 5	[1]	
valueDataType	Specifies the data type of the value needed for the interpretation of the value data	String as described in 4.1.2.9	[0..1]	"xsd:integer"

4.1.2.6 MultiValuedDataElement

The MultiValuedDataElement is a list which contains similar objects of the same data type.

It contains at least one or many DataElements. This class inherits the attributes *elementId* and *dictionaryReference* from the abstract DataElement class and adds the following specific attributes (see Table 4).

Table 4 — Definition of *MultiValuedDataElement* attributes

Name	Description	Datatype	Cardinality	Example
valueDataType	Specifies the data type of the value needed for the interpretation of the value data	String as described in 4.1.2.9	[0..1]	"xsd:integer"
[DataElement]	Data elements of the same type within the multi valued data element	DataElement (abstract) as per 4.1.2.3	[0..*]	

[DataElement] is a placeholder for additional content or other attributes of the MultiValuedDataElement.

4.1.2.7 RelatedResource

The *RelatedResource* is the semantic representation of any instance of a document or certificate that can be required to be included into a digital product passport instance. This class inherits the attributes *elementId* and *dictionaryReference* from the abstract *DataElement* class and adds the following specific attributes (see Table 5).

Table 5 — Definition of RelatedResource attributes

Name	Description	Datatype	Cardinality	Example
contentType	The IANA Media Type/MIME type	String	[1]	“application/pdf”
url	The link to the resource	String formatted as URL according to RFC 3986 [5]	[1]	
language	The language of the resource, if applicable	String formatted as localized language tag with: two (2) characters language code as in ISO 639:2023 [6] (e.g “en”), and two (2) characters country code as in EN ISO 3166-1:2020 [7]	[0..1]	“en-GB” “de-DE” “de-AT”
resourceTitle	Human readable title of the resource	String	[0..1]	“Maintenance manual”

Additionally, it can include other attributes if required by the relevant legal acts.

4.1.2.8 MultiLanguageDataElement

4.1.2.8.1 General

The *MultiLanguageDataElement* represents a language dependent data point whose value varies with respect to the language associated with it. It contains at least one or many *MultiLanguageValues* (see 4.1.2.8.2). This class inherits the attributes *elementId* and *dictionaryReference* from the abstract *DataElement* class.

4.1.2.8.2 MultiLanguageValue

The *MultiLanguageValue* represents a language dependant value. It includes the following metadata attributes (see Table 6).

Table 6 — Attributes of the MultiLanguageValue object

Name	Description	Datatype	Cardinality	Example
value	Contains the data of the value in the MultiLanguageValue	String	[1]	“a product description”

Name	Description	Datatype	Cardinality	Example
language	Specifies the language of the value within the MultiLanguageValue	String formatted as localized language tag with: two (2) characters language code as in ISO 639:2023 [6] (e.g "en"), and two (2) characters country code as in EN ISO 3166-1:2020 [7]	[1]	"en-gb" "de-de" "de-at"

4.1.2.9 Value data type

The *valueDataType* should allow XSD data types conformant to W3C XML Schema [8] and [9] except:

— the XSD data types that represent a list of values within a single whitespace-separated string (i.e. ENTITIES, IDREFS, and NMTOKENS) shall not be supported.

— XSD string derived built-in types shall not be supported, and

— the following XSD primitive types shall not be supported: NOTATION, QName.

— **base64 binary may be supported**

4.2 Data model requirements to manage DPP changes

During its lifetime, the digital product passport may undergo changes that impact the data points to be declared, like for example the environmental performance. These changes could lead to a modification of the DPP initially published for a product, either through a change of an existing property or document, or through the addition of information or a property not in the initial instance. To be properly archived, every digital product passport lifetime event resulting into a change to a DPP should be kept track of. Data include for example:

— identifier (e.g. digitalProductPassportId, uniqueProductIdentifier);

— timestamp (when the change occurred);

— actor as per EN 18239[1] [10] (who made the change);

— changed properties (specific properties modified).

Any changes to a DPP shall be logged and archived according to EN 18221 requirements. Delegated acts can define to what extent.

4.3 Model for repositories for the data dictionaries

To enable cross-sector semantic interoperability for digital product passport, while avoiding constraints on how DPP information may be defined and ensuring openness to different technologies (or technological approaches), a decoupled approach shall be applied. In this approach, data points and their definitions are managed independently.

Each data point should be associated with a machine-readable semantic definition stored in a repository. The governance of such repository should be regulation-based, but shall allow for the inclusion of definitions beyond regulatory requirements where appropriate. Each definition shall have a unique identifier and shall occur only once within a repository to enable unambiguous referencing. This also allows for mapping between definitions contained in different semantic catalogues (repositories).

This approach contributes to long-term availability and consistency of definitions, supports multilingual use, and facilitates the management of controlled value lists (e.g. non-numeric values of product properties). It also enables leaner DPP instances by reducing the volume of embedded semantic information. In the DPP semantic model ~~4.1.2.1~~, semantic definitions are referenced using the attribute *dictionaryReference*. This attribute links a given DataElement to its corresponding definition in the repository.

5 Technical interoperability

5.1 General

The definitions in this clause enable the efficient employment of the semantic representation defined in Clause 4 by notified actors, systems and processes involved with the DPP.

5.2 Serialization syntax

5.2.1 General

The serialization of data with standardized data structures shall follow the model provided in Clause 4. The data syntaxes are defined in EN 18216 [11].

To serialize the entities defined in Clause 4, the following guidelines shall be adopted.

For compressed serialization in JSON see examples in 5.2

For expanded full serialization in JSON see examples Annex A.

For serialization in XML see examples Annex B.

5.2.2 Naming conventions and built-in types

The naming conventions adopted are:

— Pascal case (PascalCase) is used for the UML Classes, example “DataElement”.

— Camel case (camelCase) is used for JSON, XML and UML attributes.

EXAMPLE “uniqueProductIdentifier”

An optimized representation of a Data Element in the conceptual model shall use the elementId as the JSON object key. The other metadata, the dictionaryReference and valueDataType are expected to be defined in external Data dictionaries (see 4.3) and therefore omitted from the operational DPP payload.

5.2.3 JSON built-in types for values

JSON native built-in types (ISO/IEC 21778 [12]) should be used to declare values if there is a specification of the data type by a data dictionary or semantic repository. If there is no such specification, then the type “*string*” in conjunction with the metadata attribute *valueDataType* may be used. The data type mappings are described in Table 7

Table 7 — Mapping of XSD to JSON Data Types

valueDataType(XSD)	JSON Type	Format / Pattern (RegEx)
xsd:string	String	N/A
xsd:boolean	Boolean	true or false
xsd:decimal	Number	A JSON Number. For high-precision values, a string may be used.

valueDataType(XSD)	JSON Type	Format / Pattern (RegEx)
xsd:integer	Number	A JSON Number with no fractional part
xsd:double, xsd:float	Number	A JSON Number.
xsd:date	String	YYYY-MM-DD (ISO 8601-1)
xsd:dateTime	String	YYYY-MM-DDThh:mm:ssZ (ISO 8601-1)
xsd:anyURI	String	A valid URI according to RFC 3986 [5]
xsd:base64Binary	String	encoded as base64

5.2.4 Serialization of the DigitalProductPassport class

A *DigitalProductPassport* class (4.1.2.1) shall be serialized in JSON for the operational data exchange as a JSON object.

The following example shows how the header of a digital product passport is serialized. These examples do not necessarily reflect real values that will be found in a digital product passport.

EXAMPLE

```
{
    //***DPP HEADER***
    "digitalProductPassportId": "https://www.example.org/dpp/1234545",
    "uniqueProductIdentifier": "https://www.example.org/1234545",
    "granularity": "Model",
    "dppSchemaVersion": "ENXXX:v1.0",
    "dppStatus": "Active",
    "lastUpdated": "2025-08-22T03:12:00Z",
    "economicOperatorId": "gxx:ppp456789",
    "facilityId": "gxx:xxx987654", // optional
    "contentSpecificationIds": [
        "prEN1234_xyz",
        "prEN5678_abc" ]
    //***END OF DPP HEADER***

    // … rest of the DPP data
}
```

5.2.5 Serialization of a DataElementCollection

A *DataElementCollection* (4.1.2.4) is serialized as a JSON object. Its fields may have any necessary combination of data types. In practice, complex nesting should be carefully considered when creating DPP requirements. These examples do not necessarily reflect real values that will be found in a digital product passport.

EXAMPLE DataElementCollection with a combination of different simple and complex types

```
{
    //...DPP HEADER...
    "performanceMetrics": { // DataElementCollection
        "maxPressure": 750.0, // SingleValuedDataElements
        "isFireResistant": true,
         "efficiencyRatings": [ // MultiValuedDataElements
             0.95,
             0.92,
              0.88
```

```
        ],
        "complexTraits": [ // additional MultiValuedDataElement
             {       // DataElementCollection
                "fieldOne": 1.0,
                "fieldTwo": "stringValueOne"
            },
             {       // DataElementCollection of same type as the first one
                "fieldOne": 2.0,
                "fieldTwo": "stringValueTwo"
            }
         ]
    }
}
```

5.2.6 Serialization of a SingleValuedDataElement

SingleValuedDataElements (4.1.2.5) are serialized as JSON «key-value pair». In case the *SingleValuedDataElement* is contained in *MultiValueDataElement*, the key-value pair may be simplified to a value of a native data type. The following examples demonstrate how *SingleValueDataElements* are used. These examples do not necessarily reflect real values that will be found in a digital product passport .

EXAMPLE 1 standalone SingleValueDataElements

```
{
    //...DPP HEADER...
    "manufacturerName": "ExampleCorp",  //type string
    "recycledContentPercentage": 25.5   //type number
}
```

EXAMPLE 2 SingleValueDataElements inside a complex data type

```
{
    //...DPP HEADER...
    "performanceMetrics": {
        "maxPressure": 750.0, // as regular key-value pairs
        "isFireResistant": true,
         "efficiencyRatings": [
             0.95, // as native values
             0.92,
            0.88
        ]
    }
}
```

5.2.7 Serialization of a MultiValuedDataElement

A *MultiValuedDataElement* (4.1.2.6) is serialized as a JSON array where all items are of the same JSON data type. This type can be a simple native type (like numbers or strings) or a complex type (like objects). When a data element is an item in a JSON array (a *MultiValueDataElement*), it does not have an explicit key. It is identified by its position (or index) within the array. These examples do not necessarily reflect real values that will be found in a digital product passport .

EXAMPLE 1 Array of native types

```
{
    //...DPP HEADER…
    "supportedVoltages":[
```

```
        110,
        220,
        240
    ]
}
```

EXAMPLE 2 Array of Objects

```
{
    //...DPPHEADER…
    "componentParts":[
        {
            "partID":"A-101",
            "material":"Steel"
        },
        {
            "partID":"B-202",
            "material":"Recycled Plastic"
        }
    ]
}
```

5.2.8 Serialization of a RelatedResource

RelatedResources (4.1.2.7) are serialized as JSON objects. These examples do not necessarily reflect real values that will be found in a digital product passport.

EXAMPLE referencing the operating instruction of a product

```
{
    "resourceTitle": "User Manual for the Smart Thermostat Pro",
    "contentType": "application/pdf",
    "url": "https://data.example.com/manuals/thermostat-pro_v2.1.pdf",
    "language": "en-GB"
}
```

5.2.9 Serialization of a MultiLanguageDataElement

MultiLanguageDataElement (4.1.2.8) is serialized as JSON array containing JSON objects, each of them serializes a *MultiLanguageValue* (4.1.2.8.2). Each *MultiLanguageValue* object contains the language dependant value and the corresponding language. These examples do not necessarily reflect real values that will be found in a digital product passport.

EXAMPLE The productDescription provided in multiple languages

```
{
    //...DPP HEADER...
    "productDescription": [
        {
            "value": "One Thing",
            "language": "en-IE"
        },
        {
            "value": "Una Cosa",
            "language": "es-ES"
        },
        {
            "value": "Une Chose",
            "language": "fr-CH"
        }
    ]
}
```

Annex A
(normative)

Expanded full data model serialized in JSON

These examples do not necessarily reflect real values that will be found in a digital product passport.

Serialization of the DigitalProductPassport class

A *DigitalProductPassport* (4.1.2.1) serialized as JSON is shown in 5.2.4

Serialization of a DataElementCollection

A *DataElementCollection* (4.1.2.4) serialized as a fully expanded JSON object is shown in Example 1:

EXAMPLE 1 DataElementCollection as fully expanded JSON object

```
{
  //...DPP HEADER...

  "elements": [
    {
      "elementId": "performanceMetrics",
      "objectType": "DataElementCollection",
      "dictionaryReference": "https://dictionary1.eu/performanceMetrics",
      "elements": [
        {
          "elementId": "maxPressure",
          "objectType": "SingleValuedDataElement",
          "dictionaryReference": "https://dictionary1.eu/maxPressure",
          "valueDataType": "xsd:float",
          "value": 2.5
        },
        {
          "elementId": "efficiencyRatings",
          "objectType": "MultiValuedDataElement",
          "dictionaryReference": "https://dictionary1.eu/efficiencyRatings",
          "valueDataType": "xsd:float",
          "elements": [
            {
              "elementId": "efficiencyRating1",
              "objectType": "SingleValuedDataElement",
              "dictionaryReference": "https://dictionary1.eu/efficiencyRating",
              "valueDataType": "xsd:float",
              "value": 0.95
            },
            {
              "elementId": "efficiencyRating2",
              "objectType": "SingleValuedDataElement",
              "dictionaryReference": "https://dictionary1.eu/efficiencyRating",
              "valueDataType": "xsd:float",
              "value": 0.92
            },
            {
              "elementId": "efficiencyRating3",
              "objectType": "SingleValuedDataElement",
              "dictionaryReference": "https://dictionary1.eu/efficiencyRating",
              "valueDataType": "xsd:float",
              "value": 0.88
```

```
            }
          ]
        }
      ]
    }
  ]
}
```

DataElementCollection with different dictionary references is shown in Example 2:

EXAMPLE 2 DataElementCollection as fully expanded JSON object with different dictionary references

```
{
    {
      "elementId": "collectionEconomicOperator",
      "objectType": "DataElementCollection",
      "dictionaryReference": "https://organizationDictionary.eu/organization",
      "elements": […]
    },
    {
      "elementId": "collectionMaterialsOfConcern",
      "objectType": "DataElementCollection",
      "dictionaryReference": "https://materialsDictionary.eu/materials/criticalMaterials",
      "elements": […]
    },
    {
      "elementId": "collectionProductSpecific",
      "objectType": "DataElementCollection",
      "dictionaryReference": "https://product-sector.eu/productDescription/productClass",
      "elements": […]
    }
```

Serialization of a SingleValuedDataElement

A *SingleValuedDataElement* (4.1.2.5) serialized as a fully expanded JSON object is shown in Example 3:

EXAMPLE 3 *SingleValuedDataElement* as fully expanded JSON object

```
{
  "elements": [
     {
       "elementId": "maxPressure",
       "objectType": "SingleValuedDataElement",
       "dictionaryReference": "https:/dictionary1/maximumPressure ",
       "valueDataType": "xsd:float",
       "value": 2.5
     },
     {
       "elementId": "recycledContentPercentage",
       "objectType": "SingleValuedDataElement",
       "dictionaryReference": "https:/dictionary1/recycledContentPercentage",
       "valueDataType": "xsd:float",
       "value": 25.5
     }
  ]
}
```

Serialization of a MultiValuedDataElement

A *MultiValuedDataElement* (4.1.2.6) serialized as fully expanded JSON object is shown in Example 4:

EXAMPLE 4 MultiValuedDataElement as a fully expanded JSON object


```
{
  "elements": [
    {
      "elementId": "efficiencyRatings",
      "objectType": "MultiValuedDataElement",
      "dictionaryReference": " https://dictionary1.eu/efficiencyRatings",
      "valueDataType": "xsd:float",
      "value": [
        {
          "elementId": "efficiencyRating1",
          "objectType": "SingleValuedDataElement",
          "dictionaryReference": " https://dictionary1.eu/efficiencyRating",
          "valueDataType": "xsd:float",
          "value": "0.95"
        },
        {
          "elementId": "efficiencyRating2",
          "objectType": "SingleValuedDataElement",
          "dictionaryReference": " https://dictionary1.eu/efficiencyRating",
          "valueDataType": "xsd:float",
          "value": "0.92"
        },
        {
          "elementId": "efficiencyRating2",
          "objectType": "SingleValuedDataElement",
          "dictionaryReference": " https://dictionary1.eu/efficiencyRating",
          "valueDataType": "xsd:float",
          "value": "0.88"
        }
      ]
    }
  ]
}
```


Serialization of a RelatedResource

A *RelatedResource* (4.1.2.7) serialized as a fully expanded JSON object is shown in Example 5:

EXAMPLE 5 RelatedResource as a fully expanded JSON object


```
{
  "elements": [
    {
      "elementId": "userManual",
      "objectType": "RelatedResource",
      "dictionaryReference": " https://dictionary1.eu/document/userManual",
      "language": "en",
      "resourceTitle": "User Manual for the Smart Thermostat Pro",
      "contentType": "application/pdf ",
      "url": "https://data.example.com/manuals/thermostat-pro_v2.1.pdf",
    }
  ]
}
```


Serialization of a MultiLanguageDataElement

A *MultiLanguageDataElement* (4.1.2.8) serialized as a fully expanded JSON object is shown in Example 6:

EXAMPLE 6 MultiLanguageDataElement as a fully expanded JSON object

```
{

  "elements": [
    {
      "elementId": "productDescription",
      "objectType": "MultiLanguageDataElement",
      "dictionaryReference": "https://dictionary1/productDescription",
      "value": [
        {
          "language": "en-IE",
          "value": "The description of a product."
        },
        {
          "language": "gr",
          "value": "Η περιγραφή ενός προϊόντος."
        },
        {
          "language": "fr-CH",
          "value": "Description d'un produit."
        }
      ]
    }
  ]
}
```

Annex B
(normative)

Data model compressed serialization in XML

These examples do not necessarily reflect real values that will be found in a digital product passport.

Serialization of the DigitalProductPassport class

A *DigitalProductPassport* (4.1.2.1) serialized as XML is shown in Example 1:

EXAMPLE 1 *DigitalProductPassport* serialized as XML

```
< dpp:DigitalProductPassport
    xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema"
    xmlns:specB = "https://ec.europa.eu/schemas/prEN5678_abc/v1.0/schema">

    <dpp:digitalProductPassportId>
       https://www.example.org/dpp/1234545</dpp:digitalProductPassportId>
    <dpp:uniqueProductIdentifier>
        https://www.example.org/1234545</dpp:uniqueProductIdentifier>
    <dpp:granularity>Model</dpp:granularity>
    <dpp:dppSchemaVersion>prEN18223:v1.0</dpp:dppSchemaVersion>
    <dpp:dppStatus>Active</dpp:dppStatus>
    <dpp:lastUpdated>2025-08-22T03:12:00Z</dpp:lastUpdated>
    <dpp:economicOperatorId>gxx:ppp456789</dpp:economicOperatorId>
    <dpp:facilityId>gxx:xxx987654</dpp:facilityId>
    <dpp:contentSpecificationIds>
        <dpp:item>prEN1234_xyz</dpp:item>
        <dpp:item>prEN5678_abc</dpp:item>
    </dpp:contentSpecificationIds>
    < !- Additional Data Elements as per specA or specB here
- > </dpp:DigitalProductPassport>
```

Serialization of a DataElementCollection

A *DataElementCollection* (4.1.2.4) serialized as XML is shown in Example 2

EXAMPLE 2 DataElementCollection serialized as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:performanceMetrics>
        <specA:maxPressure>750.0</specA:maxPressure>
        <specA:isFireResistant>true</specA:isFireResistant>
        <specA:efficiencyRatings>
            <specA:item>0.95</specA:item>
            <specA:item>0.92</specA:item>
            <specA:item>0.88</specA:item>
        </specA:efficiencyRatings>
        <specA:complexTraits>
            <specA:item>
                <specA:fieldOne>1.0</specA:fieldOne>
                <specA:fieldTwo>stringValueOne</specA:fieldTwo>
            </specA:item>
            <specA:item>
```

```
                <specA:fieldOne>2.0</specA:fieldOne>
                <specA:fieldTwo>stringValueTwo</specA:fieldTwo>
            </specA:item>
        </specA:complexTraits>
    </specA:performanceMetrics>
</dpp:DigitalProductPassport>
```

Serialization of a SingleValuedDataElement

A SingleValuedDataElement (4.1.2.5) serialized as XML is shown in Example 3 and Example 4:

EXAMPLE 3 Standalone *SingleValuedDataElement* as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:manufacturerName>ExampleCorp</specA:manufacturerName>
    <specA:recycledContentPercentage>25.5</specA:recycledContentPercentage>
</dpp:DigitalProductPassport>
```

EXAMPLE 4 *SingleValuedDataElement* inside a complex data type as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:performanceMetrics>
        <specA:maxPressure>750.0</specA:maxPressure>
        <specA:isFireResistant>true</specA:isFireResistant>
        <specA:efficiencyRatings>
            <specA:item>0.95</specA:item>
            <specA:item>0.92</specA:item>
            <specA:item>0.88</specA:item>
        </specA:efficiencyRatings>
    </specA:performanceMetrics>
</dpp:DigitalProductPassport>
```

Serialization of a MultiValuedDataElement

A *MultiValuedDataElement* (4.1.2.6) serialized as XML is serialization shown in Example 5 and Example 6:

EXAMPLE 5 MultiValuedDataElement (Array of native types) as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:supportedVoltages>
        <specA:item>110</specA:item>
        <specA:item>220</specA:item>
        <specA:item>240</specA:item>
    </specA:supportedVoltages>
</dpp:DigitalProductPassport>
```

EXAMPLE 6 MultiValuedDataElement (Array of objects) as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
```

```
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:componentParts>
        <specA:item>
            <specA:partID>A-101</specA:partID>
            <specA:material>Steel</specA:material>
        </specA:item>
        <specA:item>
            <specA:partID>B-202</specA:partID>
            <specA:material>Recycled Plastic</specA:material>
        </specA:item>
    </specA:componentParts>
</dpp:DigitalProductPassport>
```

Serialization of a RelatedResource

A *RelatedResource* (4.1.2.7) serialized as XML is shown in Example 7:

EXAMPLE 7 RelatedResource as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:userManual>
        <specA:resourceTitle>
          User Manual for the Smart Thermostat Pro
        </specA:resourceTitle>
        <specA:contentType>application/pdf</specA:contentType>
        <specA:url>https://data.example.com/manuals/thermostat-pro_v2.1.pdf</specA:url>
        <specA:language>en-GB</specA:language>
    </specA:userManual>
</dpp:DigitalProductPassport>
```

Serialization of a MultiLanguageDataElement

A *MultiLanguageDataElement* (4.1.2.8) serialized as XML is shown in Example 8:

EXAMPLE 8 MultiLanguageDataElement as XML

```
< dpp:DigitalProductPassport xmlns:dpp = "https://standards.cen.eu/dpp/18223/v1.0/schema"
    xmlns:specA = "https://ec.europa.eu/schemas/prEN1234_xyz/v1.0/schema">
    < !- Begin DPP Header Elements - >     < !- ... End DPP Header Elements
- >     <specA:productDescription>
        <specA:MultiLanguageValue>
            <specA:value>One Thing</specA:value>
            <specA:language>en-IE</specA:language>
        </specA:MultiLanguageValue>
        <specA:MultiLanguageValue>
            <specA:value>Una Cosa</specA:value>
            <specA:language>es-ES</specA:language>
        </specA:MultiLanguageValue>
        <specA:MultiLanguageValue>
            <specA:value>Une Chose</specA:value>
            <specA:language>fr-CH</specA:language>
        </specA:MultiLanguageValue>
    </specA:productDescription>
</dpp:DigitalProductPassport>
```

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of (EU) Regulation 2024/1781 aimed to be covered

This European Standard has been prepared under a Commission's standardization request *M/604 Amd 1* to provide one voluntary means of conforming to essential requirements of (EU) Regulation 2024/1781 of 13 June 2024 of the European Parliament and of the Council of the European Union establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC.

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding essential requirements of that Regulation and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Articles of (EU) Regulation 2024/1781

Essential requirements of (EU) Regulation 2024/1781	Clause(s)/subclause(s) of this EN	Remarks/Notes
Article 7(8)	4.1.2.8, 4.1.2.8.2	
Article 10.1.d	4.1	
Article 10.1.f	4.1.2	
Article 11.a	4, 5	
Article 11.d	4.1.2.7	

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the products falling within the scope of this standard.

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